

Lecture 2: Planned Notes

Thursday, May 8, 2025

9:26 AM

Review:

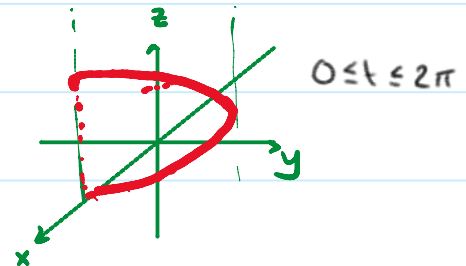
Parametric representation

$$\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}, \quad a \leq t \leq b$$

e.g. Ellipse:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \leftrightarrow \quad \vec{r}(t) = \begin{bmatrix} a \cos t \\ b \sin t \end{bmatrix}$$

e.g. Circular Helix: $\vec{r}(t) = \begin{bmatrix} a \cos t \\ a \sin t \\ ct \end{bmatrix}$



Velocity & Acceleration: $\vec{v} = \vec{r}'(t)$, $\vec{a} = \vec{r}''(t) = \vec{v}'(t)$

Speed: $|\vec{v}| = |\vec{r}'(t)| = \sqrt{\vec{r}' \cdot \vec{r}'}$

Tangent line: $q(\omega) = \vec{r}_0 + \omega \vec{v}_0$

Tangential & Normal Acceleration

$$\vec{a}_{\text{tan}} = \left(\frac{\vec{a} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \right) \vec{v}$$

$$\vec{a}_{\text{norm}} = \vec{a} - \vec{a}_{\text{tan}}$$